Roll no..............................

Name of the Course: B. Tech, CE

Semester: III

Name of the Paper: Geomatics Engineering

Paper Code: TCE 302

Time: 3 Hour

Maximum Marks: 100

Note:

(i) All questions are compulsory.
(ii) Answer any two sub question among a, b & C in each main question
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1.	**(20 Marks), CO1**
a. Discuss how the constructional features, observation techniques, and practical uses of the surveyor's compass vary from those of the prismatic compass.	
b. With the help of examples, describe the role and necessity of main survey stations, base line, and tie or check lines in chain surveying	
c. Compute the whole circle bearing of a line whose latitude and departure are given as +78 m and –45.1 m respectively.	
Q2.	(20 Marks), **CO2**
a. Discuss trilateration method in surveying. Analyze its advantages and disadvantages.	
b. Compute the true length of a 250 m line measured with a 20 m chain that was 10 cm too long	
c. Explain with diagrams the concept of triangulation, and evaluate its purpose in establishing control networks.	
Q3.	(20 Marks), **CO3**
a. Explain different sources of errors in levelling? How are they eliminated?	
b. The following consecutive readings were taken with a dumpy level 6.21. 4.92, 6.12. 8.42. 9.81, 6.63, 7.91, 8.26, 9.71, 10, 21 The level was shifted after 4th, 6th and 9th readings. The reduced level at first point was 100 ft. Rule out a page of your answer-book as a level field book and fill all the columns. Use collimation system and apply the usual arithmetical check	
c. Explain Contour and describe characteristics of contours with the help of sketches.	
Q4.	(20 Marks), **CO4**
a. Discuss two-point problem. How is it solved?	
b. Explain different sources of errors in plane tabling. How are they eliminated?	
c. Discuss Digital elevation model along with the creation methods. Also mention key application areas.	
Q5.	(20 Marks), **CO5**
a. Explain the difference between random errors and systematic errors, providing a practical example for each in surveying	
b. Explain the fundamental principle of the method of least squares and its purpose in survey adjustment computations	
c. A horizontal angle θ is measured by four different surveyors with varying quality of instruments and procedures. The recorded values and their assigned weights (based on the number of observations or instrument quality) are:	

36°30′ (Weight 2)
36°00′ (Weight 1)
35°30′ (Weight 4)
36°30′ (Weight 2) Calculate the most probable value of the angle θ.